

Adaptive Explanation Obfuscation via Reinforcement Learning

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Abstract—Artificial intelligence (AI) models typically operate as opaque black boxes, producing outputs without revealing the reasoning behind them. This lack of transparency is undesirable, as clients submitting queries often require insight into how results are obtained. Explainable AI (XAI) addresses this need by providing interpretable explanations of model behavior. However, the very process of explanation can expose vulnerabilities: a malicious client may exploit explanations to extract sensitive information about the underlying model. Such model extraction attacks pose a significant threat to model owners, jeopardizing intellectual property and research investments. Consequently, developing robust defenses against these attacks has become a critical area of research and in this work, we present solution to remedy such threats. The solution introduces a reinforcement learning (RL) agent that functions as a protective layer around an XAI model, dynamically adjusting the level of explanation provided based on ongoing interactions. The problem is structured as a continuous adversarial min-max game: an adversary attempts to reconstruct a surrogate model using the explanations given, while the RL agent actively obfuscates the explanations to prevent successful extraction. The system aims to satisfy legitimate user needs while penalizing the agent if the adversary successfully predicts or reconstructs the model. Over time, the agent learns to adapt its strategy, minimizing risks while maintaining the semantic usefulness of explanations for authorized users across varying model architectures.

Index Terms—Explainable AI (XAI), Reinforcement Learning (RL), Model Extraction Attacks, Privacy, Obfuscation.

I. INTRODUCTION

Explainable Artificial Intelligence (XAI) focuses on making complex models and their predictions interpretable, providing insights into the reasoning processes behind machine learning outputs. However, while offering explanations assists legitimate users in comprehending model behavior, it simultaneously creates potential vulnerabilities that adversaries may leverage to replicate models.

A model extraction attack occurs when an adversary attempts to replicate a black box model. If an attack utilizes explanations, it is called an explanation-aware model extraction attack (XaMEA). In such attacks, the adversary sends a massive volume of crafted queries to reveal information about the target model's behavior. The explanations consist of additional information such as feature importance or confidence scores, which the attacker uses to match both the outputs and explanations of a surrogate model to those of the target model.

Therefore, when accompanied by explainability, even highly complex models remain vulnerable to extraction attacks.

To address these challenges, we introduce an approach that integrates a reinforcement learning (RL) agent with XAI to dynamically enhance privacy and security without strictly compromising the explanation quality for regular users.

II. RELATED WORK

To contextualize our dynamic obfuscation approach, we review the literature across four key domains: model extraction vulnerabilities, the privacy-explainability trade-off, static defensive mechanisms, and the application of reinforcement learning in adversarial defense.

A. Vulnerabilities in Machine Learning (Model Extraction)

Standard machine learning models exposed via inference APIs are highly susceptible to model extraction (or model stealing) attacks [1], [2]. In these scenarios, adversaries systematically query the prediction API and utilize the resulting input-output pairs to train a functionally equivalent surrogate model [3]. Recent advances have demonstrated that these attacks can achieve both high accuracy and high fidelity representations of the target model [4]. This vulnerability extends beyond simple tabular classifiers, threatening complex sequential recommenders [5] and large language models [6]. Comprehensive surveys of the field emphasize that modern extraction methodologies are becoming increasingly query-efficient, drastically reducing the time and cost required for an adversary to steal proprietary intellectual property [7].

B. The Privacy-Explainability Trade-off and XaMEA

The deployment of XAI methodologies, such as SHAP [8], LIME [9], and Integrated Gradients [10], has become a standard practice for establishing algorithmic transparency. However, the literature increasingly recognizes a strict privacy-explainability trade-off [11]. Explanations inherently leak gradient information and expose local decision boundaries [12]. Milli et al. [13] theoretically and empirically demonstrated that gradient-based explanations allow for rapid, query-efficient model reconstruction.

This inherent vulnerability spawned the development of Explanation-Aware Model Extraction Attacks (XaMEA) [14]. By exploiting feature importances alongside standard outputs,

attackers can accelerate the extraction process. Attack frameworks such as AUTOLYCUS [15] leverage model-agnostic tools like LIME and SHAP to infer decision boundaries with minimal queries, while similar architectures like Megrez [16] effectively target Deep Neural Networks. Furthermore, Shokri et al. [17] established that detailed explanations also exacerbate membership inference risks, confirming that transparency tools uniquely expose models to adversarial exploitation and manipulation [18].

C. Current Defensive Strategies and Their Limitations

Mitigating model extraction traditionally relies on static defense mechanisms. Common techniques include Top- k truncation [5], which limits the informational bandwidth by restricting output dimensions, and prediction generalization, which obscures outputs through deliberate rounding [1]. More proactive defenses attempt to inject controlled, randomized noise into the API predictions [11], [19], often drawing upon differential privacy (DP) frameworks [20] to provide mathematical privacy guarantees for interpretable models [21].

Other state-of-the-art approaches, such as PRADA [22] and adaptive misinformation frameworks [23], [24], analyze incoming query distributions to detect anomalous behavior and intentionally mislead attackers. However, when these static noise or truncation defenses are applied to XAI outputs, they inflict severe utility loss [11]. Static defenses cannot dynamically balance the conflicting needs of maintaining high explanation fidelity for benign users while securing the model against active adversaries.

D. Reinforcement Learning in Adversarial Defense

To effectively combat adaptive adversaries, defensive mechanisms must also be adaptive and unpredictable [25]. Reinforcement Learning (RL) has emerged as a powerful paradigm for addressing complex, dynamic challenges in cybersecurity [26]. Foundational algorithms like Deep Q-Networks (DQN) [27] and Proximal Policy Optimization (PPO) [28] have been successfully adapted for real-time threat detection and network intrusion mitigation [29].

In the specific context of adversarial machine learning, frameworks like Robust Adversarial Reinforcement Learning (RARL) [30] formulate security as a continuous min-max game. This forces the RL agent to learn robust behavioral policies against active perturbations. By extending this game-theoretic RL approach to XAI obfuscation, our system overcomes the core limitations of static privacy mechanisms, allowing the model to dynamically optimize the trade-off between interpretability and security on a per-interaction basis.

III. PROPOSED SYSTEM ARCHITECTURE

The system consists of an inference provider who manages the ML model along with its associated explainability module. The RL agent acts as a protective mechanism that decides how much information to conceal or modify in an explanation.

As illustrated in Fig. 1, the high-level components of the system are:

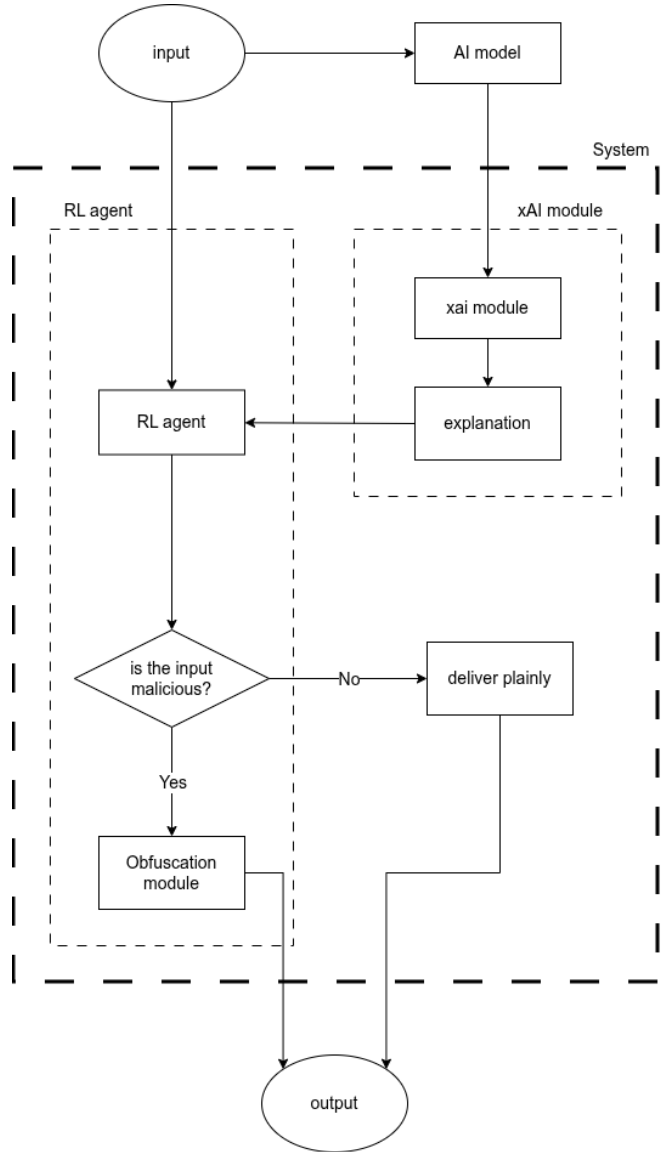


Fig. 1. High-level overview of the system architecture.

- **AI Model:** A model hosted and served by the inference provider. In our evaluations, we utilize both Decision Tree Ensembles (XGBoost) and Deep Neural Networks (DNN) to demonstrate architectural generalizability.
- **Input:** The query submitted for processing.
- **RL agent:** Determines how much information to conceal in an explanation to limit model extraction while preserving utility. We utilize a Proximal Policy Optimization (PPO) algorithm for this agent.
- **Obfuscation module:** Intentionally obfuscates explanations to prevent attackers from learning too much about the model.
- **XAI module:** Generates baseline explanations for the outputs of the model using frameworks such as SHAP or LIME.

The workflow operates as follows (see Fig. 2):

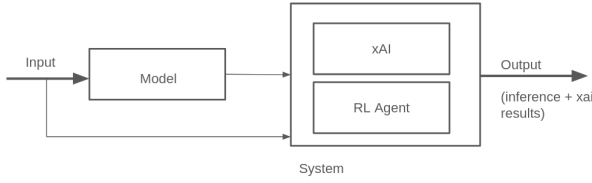


Fig. 2. System workflow: inference model output is explained by the XAI module and selectively obfuscated by the RL agent before returning results.

- 1) An input query is submitted to the system.
- 2) The input is processed on the model, generating an output.
- 3) The input is simultaneously sent to the RL agent for analysis.
- 4) The RL agent examines the input's relation to historical queries and determines the degree of obfuscation required.
- 5) The XAI module receives the output and generates a baseline explanation, which is sent to the RL agent.
- 6) The RL agent decides whether to apply obfuscation to the explanation via the obfuscation module.
- 7) The processed output and the (potentially obfuscated) explanation are returned to the user.

IV. GAME DESIGN FOR THE RL AGENT

A. Reinforcement Learning Configuration

Algorithm Selection: To drive the dynamic obfuscation module, we employ the Proximal Policy Optimization (PPO) algorithm [28]. PPO was selected over alternative reinforcement learning algorithms, such as Deep Q-Networks (DQN) or Soft Actor-Critic (SAC), due to its inherent stability and compatibility with continuous action spaces. Because our obfuscation intensity a_t is a continuous variable defined in the range $[0, 1]$, value-based methods like DQN, which require discrete action spaces, are unsuitable without arbitrary discretization. While SAC accommodates continuous actions, PPO's clipped surrogate objective function prevents destructively large policy updates, ensuring stable and monotonic improvement during the delicate min-max adversarial training process.

State-Space Formulation and Network Architecture: The RL agent must make contextual decisions based on the current query and the history of adversarial interactions. To achieve this efficiently without the computational overhead of processing full sequential arrays, the state-space s_t embeds the temporal context via a history ratio. Specifically, the observation is formulated as the concatenation of the flattened current feature query x_t and the normalized step counter:

$$s_t = \left[x_t \parallel \frac{t}{T_{max}} \right] \quad (1)$$

Where t is the current query step and T_{max} represents the maximum allowed queries in a session. This combined vector is processed by the PPO agent, which is parameterized using a standard Multi-Layer Perceptron (MLP) architecture. The MLP network acts as both the actor, outputting the continuous

action a_t , and the critic, estimating the state-value function to guide policy updates.

Convergence Behavior: During training, the PPO agent demonstrated robust convergence. Operating over 50,000 environmental timesteps with a learning rate of 3×10^{-4} , the agent successfully stabilized its defensive policy. Empirical learning curves derived from the environment's rolling episode rewards indicate that the agent steadily maximized its reward function, confirming that PPO effectively managed the volatile signals generated by the adversary's simultaneously evolving surrogate model.

The RL agent is trained using a game-based framework designed to enhance model security while maintaining explanation utility. The setup simulates a continuous adversarial scenario, as depicted in Fig. 3.

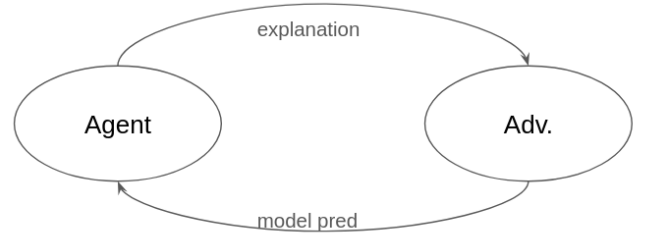


Fig. 3. Game-based training loop: the RL agent obfuscates explanations; the adversary trains a surrogate model, and rewards/penalties are assigned based on security and explanation utility.

B. Min-Max Game Formulation

Step 1: State at time t

The state s_i at time t is the history of queries received from a specific user, where x_i is the query and y_i is the output.

$$s_i = \{(x_1, y_1), (x_2, y_2), \dots, (x_i, y_i)\} \quad (2)$$

Step 2: RL Agent Action

The RL agent determines an obfuscation function f_{obfus} to apply to the true explanation E_{true} . The action a_t is a continuous variable $a_t \in [0, 1]$ representing the intensity of obfuscation. The applied transformation is defined as:

$$E_{output} = (1 - a_t)E_{true} + a_t \cdot \epsilon \quad (3)$$

Where ϵ represents Gaussian noise scaled by the standard deviation of the true explanations ($\epsilon \sim \mathcal{N}(0, \sigma_E^2)$). If $a_t = 0$, $E_{output} = E_{true}$ (full explanation). If $a_t = 1$, E_{output} relies entirely on the generated noise (full obfuscation).

Step 3: Adversary Action

The adversary leverages E_{output} to construct a surrogate model S targeting the original model M . The adversary minimizes the extraction loss $L_{extract}$:

$$\min L_{extract}(S(x), M(x) | E_{output}) \quad (4)$$

Step 4: Reward Function

The reward function R_t balances security (maximizing adversary loss) and utility (minimizing explanation deviation).

$$R_t = \lambda \cdot L_{extract}(S, M) - \mu \cdot \|E_{true} - E_{output}\| \quad (5)$$

Where λ and μ are non-negative scalar hyperparameters weighing the importance of model security and explanation fidelity, respectively.

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

To evaluate the proposed methodology and demonstrate its structural generalizability, we implemented two distinct machine learning pipelines:

- 1) **Adult Income (XGBoost + SHAP):** A black-box XGBoost target model providing interpretability via SHAP TreeExplainer.
- 2) **Credit Card Fraud (DNN + LIME):** A Deep Neural Network target model providing interpretability via LIME (Local Interpretable Model-agnostic Explanations).

In both environments, the adversary was designed as an adaptive attacker aiming to replicate the target model using a surrogate Multi-Layer Perceptron (MLP) Classifier configured with two hidden layers of 64 and 32 neurons. The RL defending agent was trained independently for each pipeline using the PPO algorithm over 50,000 timesteps.

B. Evaluation Metrics

We benchmark the defensive policies across three primary metrics to assess the balance between adversarial security and user interpretability:

1. Average Extraction Loss ($L_{extract}$): Measures the cross-entropy loss of the adversary’s surrogate model. Higher values indicate successful defense against extraction.

2. Average Utility Loss (L2 Distortion): The normalized L2 norm between the true explanation and the obfuscated output, representing the raw geometric distortion.

3. Semantic Interpretability Metrics: To ensure the obfuscation policy does not destroy the inherent meaning of the explanations, we evaluate two semantic utility metrics:

- **Spearman’s Rank Correlation (ρ):** Measures the monotonic relationship between the feature importance rankings of the original explanation E_{true} and the obfuscated explanation E_{output} . A higher ρ indicates that the relative hierarchy of feature importance is preserved.
- **Top- k Feature Agreement:** Calculates the intersection of the k most critical features before and after obfuscation, defined as $\frac{|S_{true} \cap S_{output}|}{k}$, where S represents the set of indices for the top k features. We report results for $k = 3$.

C. Results and Baselines

The trained PPO agents were benchmarked against multiple static defense strategies prevalent in the literature: Top- k truncation ($k = 3$), Gaussian Noise addition (level=0.5), Precision Reduction (2 decimals), and Random Subset omission ($p = 0.5$).

As demonstrated in Table I, static methods struggle to balance the tradeoff effectively on the SHAP pipeline. While Gaussian Noise and Random Subset methods exhibit higher security, they inflict severe utility loss (0.997017 and

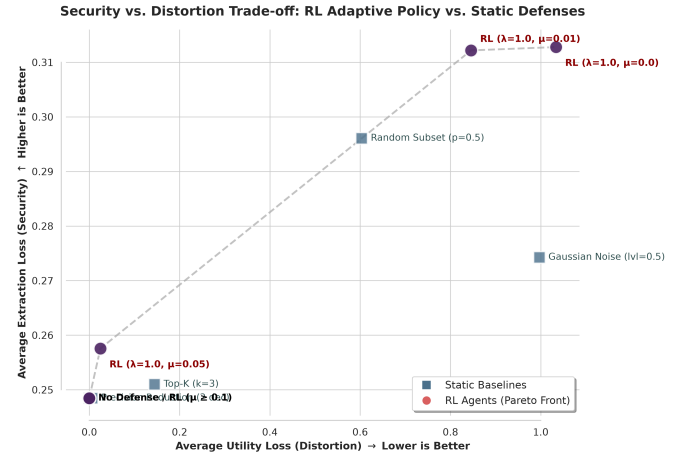


Fig. 4. Security versus Distortion Pareto Front

0.603572, respectively), rendering the explanations functionally useless for benign users. In contrast, the RL Agent ($\mu = 0.05$) successfully increases security (0.257550) over the baseline (0.248464) while maintaining an exceptional explanation fidelity (Top-3 Agreement ≈ 0.98).

The evaluation on the Credit Card Fraud dataset (Table II) further demonstrates the severe limitations of static defenses when applied to LIME explanations. While Top- k Truncation ($k = 3$) maintains perfect Top-3 Agreement, it completely disrupts the overall feature ranking ($\rho \approx 0.52$). Conversely, Precision Reduction preserves general ranking ($\rho \approx 0.82$) but loses over half of the most critical top-3 features (Agreement ≈ 0.48). Our adaptive policy ($\lambda = 1.0, \mu = 0.01$) successfully navigates this trade-off, maintaining high rank correlation ($\rho \approx 0.94$) and preserving the most critical features (Top-3 Agreement ≈ 0.89) while defending against the extraction.

VI. DISCUSSION AND PRACTICAL DEPLOYMENT

A. Addressing Computational Overhead

A primary concern when integrating adaptive defense mechanisms into production machine learning pipelines is the introduction of unacceptable latency during real-time inference. Traditional reinforcement learning training is computationally intensive; however, the deployment phase of our proposed architecture introduces negligible overhead.

The trained PPO actor is parameterized as a lightweight Multi-Layer Perceptron (MLP). During inference, calculating the obfuscation intensity a_t requires only a single forward pass through this shallow neural network. The computational bottleneck in the proposed architecture remains the generation of the baseline XAI explanation (e.g., calculating SHAP or LIME values), not the execution of the RL defensive policy. Therefore, the dynamic obfuscation layer can be deployed as an API gateway proxy without violating strict latency service-level agreements (SLAs).

TABLE I
SECURITY VS. DISTORTION TRADE-OFF: ADULT INCOME DATASET (XGBOOST + SHAP)

Strategy	Extraction Loss $\uparrow$	Utility Loss $\downarrow$	Spearman's ρ $\uparrow$	Top-3 Agreement $\uparrow$
<i>Static Baselines</i>				
No Defense	0.248464	0.000000	1.000000	1.000000
Precision Reduction (2 dec)	0.248468	0.006274	0.997749	0.994000
Top- k ($k = 3$)	0.250986	0.144675	0.941124	1.000000
Gaussian Noise (lvl=0.5)	0.274254	0.997017	0.611314	0.731333
Random Subset ($p=0.5$)	0.296077	0.603572	0.564830	0.552667
<i>RL Adaptive Agents ($\lambda = 1.0$)</i>				
RL ($\mu \geq 0.1$)	0.248464	0.000000	1.000000	1.000000
RL ($\mu = 0.05$)	0.257550	0.024615	0.986171	0.986667
RL ($\mu = 0.01$)	0.312179	0.845807	0.565143	0.710667
RL ($\mu = 0.0$)	0.312788	1.034102	0.469371	0.671333

TABLE II
SECURITY VS. DISTORTION TRADE-OFF: CREDIT CARD FRAUD DATASET (DNN + LIME)

Strategy	Extraction Loss $\uparrow$	Utility Loss $\downarrow$	Spearman's ρ $\uparrow$	Top-3 Agreement $\uparrow$
<i>Static Baselines</i>				
No Defense	0.991585	0.000000	1.000000	1.000000
Precision Reduction (2 dec)	0.983225	0.491624	0.828449	0.485333
Top- k ($k = 3$)	0.987773	0.770182	0.528731	1.000000
Gaussian Noise (lvl=0.5)	0.984138	0.662956	0.807614	0.527333
Random Subset ($p=0.5$)	0.985825	0.708359	0.644454	0.462667
<i>RL Adaptive Agents ($\lambda = 1.0$)</i>				
RL ($\mu = 0.05$)	0.987885	0.049346	0.977625	0.965333
RL ($\mu = 0.01$)	0.983497	0.130573	0.946349	0.898667
RL ($\mu = 0.0$)	0.990951	0.216321	0.915109	0.838667

B. Dynamic Threat Posturing

The ablation study demonstrates that a single RL architecture can accommodate diverse security postures. In a practical deployment, a centralized security orchestration system could dynamically swap the active RL agent weights based on the current global threat level. During periods of normal operation, the system can utilize a high-utility agent ($\mu = 0.05$). If an intrusion detection system (IDS) flags an ongoing distributed XaMEA attack, the gateway can instantly pivot to a high-security agent ($\mu = 0.01$), actively stifling the extraction attempt while still providing baseline, albeit noisy, interpretability.

VII. ADVANTAGES OF THE PROPOSED SOLUTION

The dynamic integration of the RL agent provides several technical benefits:

- **Enhanced Security:** Mitigates model extraction attacks by limiting explanation detail during suspicious activity.

- **Adaptive Explanations:** Continuously monitors query patterns and adapts the detail level based on historical interactions.
- **Balanced Utility:** As evidenced by the empirical benchmarks, authorized users receive high-quality insights with negligible semantic distortion, while adversaries are successfully stifled.
- **Robustness:** The RL agent adapts over time, effectively countering evolving attack strategies across varied architectures (Tree Ensembles and DNNs) and XAI methods (SHAP and LIME).

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